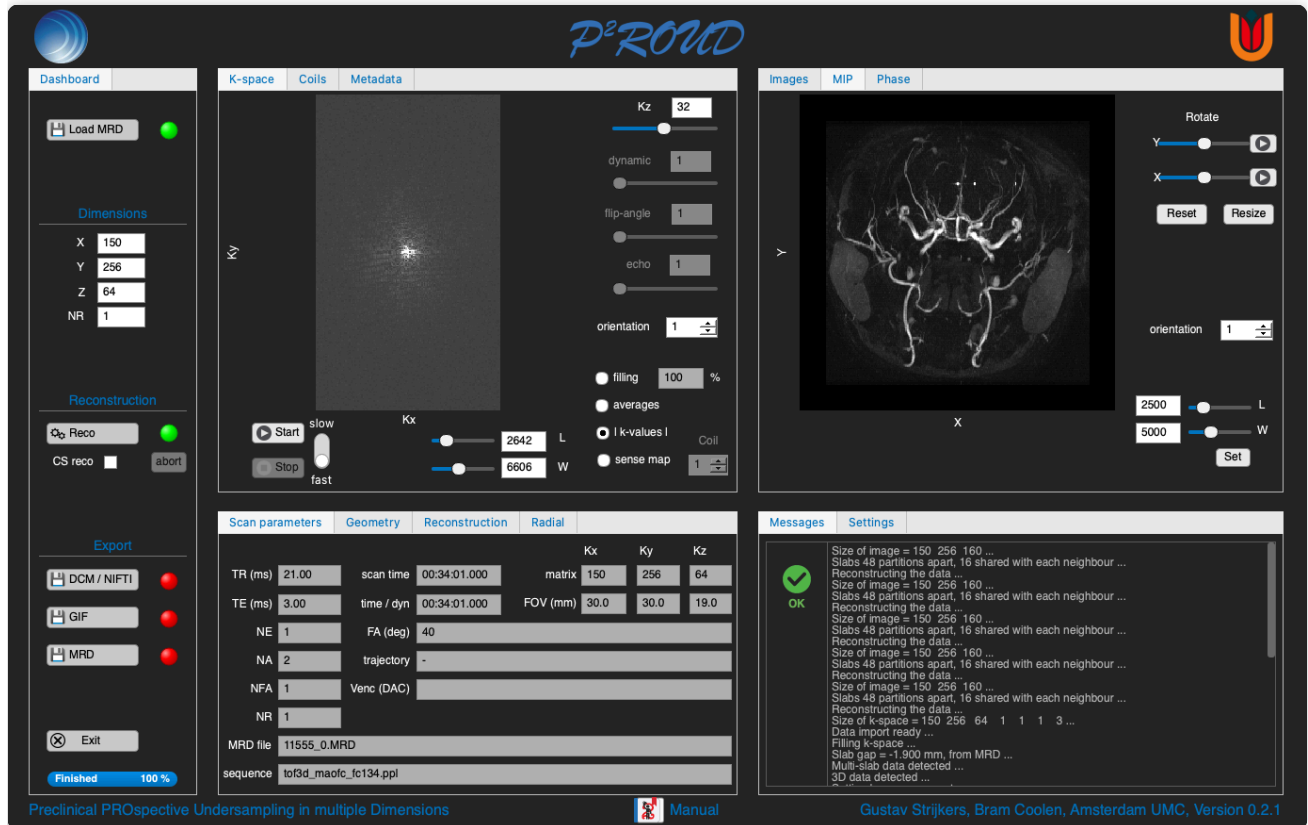




## Table of contents

1. Overview
2. Contact
3. Disclaimer
4. Installation
5. Running the software
6. Basic operation
7. Credits



## 1. Overview

P2ROUD = Preclinical PROspective Undersampling in multiple Dimensions.

P2ROUD is an application with a graphical user interface to reconstruct MRI data acquired with an MR Solutions preclinical MRI system, in particular data with k-space undersampling in one or more dimensions. Besides Cartesian 2D and 3D data it reconstructs 2D radial, 3D UTE and ZTE scans, and 2D EPI scans for diffusion and fMRI, which it can export as a BIDS dataset.

P2ROUD is a stand-alone application: everything it uses is inside it, and nothing else has to be installed.

## 2. Contact

P2ROUD and this manual are written by:

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## 3. Disclaimer

This is a beta version. The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers or Bram Coolen.



## 4. Installation

### macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

P2ROUD comes as a disk image, `P2ROUD-0.2.2-macOS-arm64.dmg`. Nothing else has to be installed: Python, `dcm2niix` and everything else the app uses are inside it.

1. Open the disk image and drag **P2ROUD** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying P2ROUD was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

Instead of step 2, the same in Terminal, once:

```
xattr -dr com.apple.quarantine /Applications/P2ROUD.app
```

The first start can take a little longer: macOS checks the app once.

### Windows (provisional)

*The Windows version has not been released yet. This section describes the installer as it is prepared, and will be corrected once it has been tested.*

Requirements: Windows 10 or 11, 64-bit.

P2ROUD comes as an installer, `P2ROUD-0.2.2-setup.exe`. It installs the app for the current user in `%LOCALAPPDATA%\Programs\P2ROUD` (no administrator rights needed; installing for all users is an option), with a Start-menu entry, an optional desktop shortcut and an uninstaller. Nothing else has to be installed.

### BART (optional)

With BART, the Berkeley Advanced Reconstruction Toolbox (<https://mricon.github.io/bart/>), the compressed sensing reconstructions run with BART. Without it, Cartesian 2D and 3D, 2D radial and 3D UTE data are reconstructed with compressed sensing by the app's own engine, based on `sigpy`, with all four regularization terms, the ESPIRiT coil maps and the density correction; it takes two to six times as long. ZTE data are then reconstructed by gridding, without compressed sensing. P2ROUD is tested with BART 1.0.00.

At its start the app writes BART's version in the message box, followed by "BART toolbox detected ..." or "WARNING: BART toolbox not available ...", and whether BART was compiled with GPU support. A BART built for the GPU is used on the GPU.

### Installing BART on macOS with MacPorts

1. Install the Xcode command line tools:

```
xcode-select --install
```



2. Install MacPorts from <https://www.macports.org/>.

3. Install the packages:

```
sudo port install fftw-3-single gcc14 libpng openblas flock gmake coreutils l
ibblocksruntime
```

4. Download and unpack BART:

```
wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
tar xvfz v1.0.00.tar.gz
cd bart-1.0.00
```

5. Build and install:

```
export PATH=/opt/local/libexec/gnubin:$PATH
CC=gcc-mp-14 gmake
sudo gmake install
```

The compiler name must match the gcc version installed in step 3: gcc14 gives gcc-mp-14, gcc15 gives gcc-mp-15.

6. Check that it works: `bart version`

## Installing BART on macOS with Homebrew

1. Install the Xcode command line tools: `xcode-select --install`

2. Install Homebrew:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

3. Install the packages BART needs:

```
brew install gcc libpng fftw openblas make llvm libomp
```

4. Download and unpack BART as above (step 4 of MacPorts).

5. Build and install:

```
CC=gcc-15 MACPORTS=0 gmake
sudo gmake install
```

The compiler name must match the version Homebrew installed; check it with `ls /opt/homebrew/bin/gcc-*`.

6. Check that it works: `bart version`

## How P2ROUD finds BART on macOS

In the normal case there is nothing to configure: `sudo make install` puts BART in `/usr/local/bin`, which is where the app looks first. It looks in these folders, in this order:

### Folder

|                                |                                                    |
|--------------------------------|----------------------------------------------------|
| <code>/usr/local/bin</code>    | where BART's own <code>make install</code> puts it |
| <code>/usr/bin</code>          |                                                    |
| <code>/opt/homebrew/bin</code> | Homebrew's                                         |
| <code>/opt/local/bin</code>    | MacPorts'                                          |



An app opened from the Finder or the Dock does not see the settings of your shell (such as `TOOLBOX_PATH` or `PATH` in your shell profile). If your `bart` is in another folder, a link to it in `/usr/local/bin` lets the app find it. In Terminal, once, with the folder your `bart` is in instead of `/path/to/bart` (Terminal asks for your password):

```
sudo mkdir -p /usr/local/bin
sudo ln -s /path/to/bart/bart /usr/local/bin/bart
```

### Installing BART on Windows 11 (provisional)

On Windows, BART runs inside the Windows Subsystem for Linux (WSL), in the default distribution; the app calls it as `wsl bart`.

1. Install WSL. Start a Windows PowerShell as administrator and run:

```
wsl --install
```

This enables the feature, installs WSL 2 and installs Ubuntu in one step. A restart is needed. The first time you start the Linux prompt you are asked to create a user account. On older systems, or if `wsl --install` is not recognized, see <https://pureinfotech.com/install-wsl-windows-11/>.

2. Update the Linux distribution (at the Linux prompt):

```
sudo apt-get update
sudo apt-get dist-upgrade
```

3. Install the packages BART needs (gcc 12 or newer):

```
sudo apt-get install make gcc libfftw3-dev liblapacke-dev libpng-dev libopenblas-dev gfortran
```

4. Download and unpack BART:

```
wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
tar xvfz v1.0.00.tar.gz
cd bart-1.0.00
```

5. Build and install, or with GPU support as below:

```
make
sudo make install
```

6. Check that it works: at the Linux prompt `bart version`, and from a Windows command prompt `wsl bart version`.

### Building BART with GPU support on Windows (provisional, highly recommended)

1. Install the NVIDIA driver in Windows, as described at <https://docs.nvidia.com/cuda/wsl-user-guide/index.html>. It is what gives WSL the GPU; no Linux display driver goes inside WSL.
2. Install the CUDA toolkit inside WSL, from <https://developer.nvidia.com/cuda-downloads> (Linux, x86\_64, WSL-Ubuntu). For CUDA 12.8:

```
wget https://developer.download.nvidia.com/compute/cuda/12.8.0/local_installers/cuda_12.8.0_570.86.10_linux.run
sudo sh cuda_12.8.0_570.86.10_linux.run
```



3. Set the environment variables, in the shell you build in: `CUDA_BASE` for the CUDA version installed, `GPUARCH_FLAGS` for the card. BART has to be built for the card's generation (look it up at <https://developer.nvidia.com/cuda-gpus>): a build for another one compiles without complaint and fails at the first calculation.

| Generation   | Cards                   | GPUARCH_FLAGS |
|--------------|-------------------------|---------------|
| Pascal       | GTX 1060, 1080, Titan X | -arch=sm_61   |
| Turing       | RTX 2060 to 2080        | -arch=sm_75   |
| Ampere       | RTX 3060 to 3090        | -arch=sm_86   |
|              | A100                    | -arch=sm_80   |
| Ada Lovelace | RTX 4060 to 4090        | -arch=sm_89   |
| Blackwell    | RTX 5070 to 5090        | -arch=sm_120  |

```
export CUDA=1
export CUDA_BASE=/usr/local/cuda-12.8
export CUDA_LIB=lib64
export GPUARCH_FLAGS="-arch=sm_61"
export PATH=$CUDA_BASE/bin:$PATH
export LD_LIBRARY_PATH=$CUDA_BASE/$CUDA_LIB:$LD_LIBRARY_PATH
```

4. Make. If the linking fails with an error about `libstdc++` or `libgcc`, point the linker at the installed gcc with the last line (its number from `ls /usr/lib/gcc/x86_64-linux-gnu/`) and make again.

```
make -j$(nproc) CUDA=1 CUDA_BASE=$CUDA_BASE CUDA_LIB=$CUDA_LIB GPUARCH_FLAGS
="$GPUARCH_FLAGS"
sudo make install
export LDFLAGS="-L/usr/lib/gcc/x86_64-linux-gnu/13 $LDFLAGS"
```

5. Test: `make utest_gpu`



## 5. Running the software

Open **P2ROUD** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows, provisional). At its start the app looks for BART, and the message box reports what it found; "Ready" says it is done.

**Newer versions.** At its start the app asks GitHub (<https://github.com/Moby1971/MRI-apps>) whether there is a newer release of it. The message box then shows the current and the newest version and, when there is a newer one, where to download it. Without a network connection it says that the version could not be checked, and the app works as usual. `checkForUpdates` in the settings file switches the check off; `versionCheckTimeout` sets how many seconds it waits for GitHub (5).

**This manual.** The **Manual** button at the bottom of the window, with the app's icon on a book, opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

**The first Load MRD dialog** opens in the scanner's MRD folder (on the scanner computer, Windows), or in the folder written in `mrddpath.txt`, a text file in the settings folder below.

### Settings and log files

P2ROUD keeps its files in a folder of its own, which it makes at its first start:

|                       |                                                          |
|-----------------------|----------------------------------------------------------|
| macOS                 | <code>~/Library/Application Support/P2ROUD/0.2.2p</code> |
| Windows (provisional) | <code>%APPDATA%\P2ROUD\0.2.2p</code>                     |

**Settings.** The settings are kept in `p2roudSettings.json`. The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds, among others, the default compressed sensing weights and iterations for each type of data, whether the EPI corrections and the ZTE registration are ticked at start, and the field strength written into DICOM headers made without the scanner's own when the scan's `.rpr` file gives none (7 T). A value that cannot be used is replaced by its standard value, and the message box says so.

**Log files.** Every session writes a log file, `p2roud-log-<date and time>.txt`, to the same folder. The log keeps every message, also those the message box does not show at the chosen message level.

**Other files in the folder.** `ktraj.txt`, the 3D UTE readout profile (see step 1); `mrddpath.txt`, when you add it; and `graddelays.txt`, when you add one, which takes the place of the app's own gradient delays.



## 6. Basic operation

The P2ROUD program operates from a single window with 5 panels.

**Panel 1: Dashboard.** The task buttons and parameters that control the reconstruction process. The dashboard tasks need to be completed from top to bottom. A green light next to a task indicates that the task has been completed; yellow, that it is busy; red, that it has not been completed yet, or failed. At the bottom of the Dashboard one progress bar shows the progress of every task: what it is doing and how far it has got ("Loading", "Reconstruction", "Export 30 of 120", "Finished").

**Panel 2: K-space / Coils / Metadata.** The raw k-space data, and the coils and their sensitivity maps. The **Metadata** tab beside them holds the record of the scan: the animal, the session and the contrast agent (step 8).

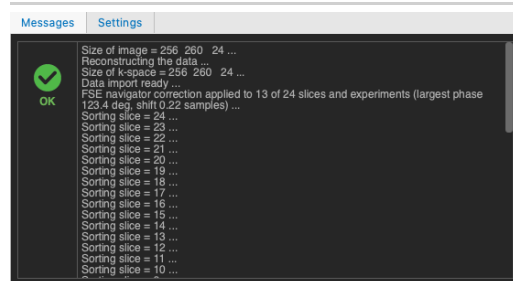
**Panel 3: Scan parameters / Geometry / Reconstruction / Radial.** The acquisition, geometry, reconstruction and radial reconstruction parameters.

**Panel 4: Images / MIP / Phase.** After reconstruction, the reconstructed images.

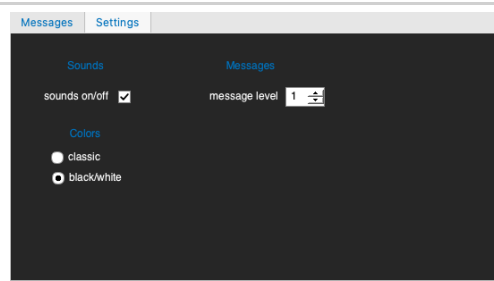
**Panel 5: Messages / Settings.** Program status and messages, newest on top. In the Settings tab the sounds can be switched on or off (**sounds on/off**), the **classic** or **black/white** colors chosen (black/white is the default), and the **message level** (1 to 3) set: how much the message box shows. The status icon shows OK, WARNING, ERROR or BUSY.



### Messages



### Settings



### Step 1: Loading data

Press **Load MRD** to load the MRD raw data file.

P2ROUD can reconstruct spin-echo, fast spin-echo, gradient-echo, MPRAGE and CINE data, multi-slab 3D, 2D radial, 3D UTE and ZTE data, and 2D EPI (diffusion and fMRI). However, the program is particularly designed to reconstruct undersampled data with a user-defined undersampling pattern (e.g. SENSE, random undersampling, pseudo-spiral or radial).

For data acquired with multiple receivers (phased-array RF coil) select one of the MRD files; the app finds the others. The relevant acquisition parameters are shown in panel 3.



## Scan parameters

| Scan parameters | Geometry        | Reconstruction | Radial       |
|-----------------|-----------------|----------------|--------------|
| TR (ms)         | 2500.00         | scan time      | 00:09:04.000 |
| TE (ms)         | 30.00           | time / dyn     | 00:09:04.000 |
| NE              | 1               | FA (deg)       | 90           |
| NA              | 2               | trajectory     | rtable.rtv   |
| NFA             | 1               | Venc (DAC)     |              |
| NR              | 1               |                |              |
| MRD file        | 17149_000_0.MRD |                |              |
| sequence        | fse3d_2702.ppl  |                |              |

## Geometry

| Scan parameters       | Geometry | Reconstruction | Radial |
|-----------------------|----------|----------------|--------|
| FOV (mm)              | 80.00    | offset X (mm)  | -0.81  |
| aspect ratio          | 0.75     | offset Y (mm)  | -4.68  |
| phase orientation     | 1        | offset Z (mm)  | -0.52  |
| slice thickness (mm)  | 1.20     | angle X (deg)  | -90    |
| slice gap (mm)        | 0.30     | angle Y (deg)  | 2      |
| slab overlap (pixels) | 1        | angle Z (deg)  | 0      |

**k-space trajectory.** There are three options to include a user-defined k-space trajectory in the FLASH (version 635e) sequence.

1. **2D data** — pe1-order = 3 ("LUT"). The k-space points for the 1st phase-encoding direction (views) are provided in the views table in the sequence. The trajectory field shows `mul_kspace_traj`.
2. **3D data** — pe1-order = 4 ("exLUT"). The k-space points for the 1st and 2nd phase-encoding directions (views and views2) are provided in a text file (\*.txt) as a list of points without comma separation, for example:

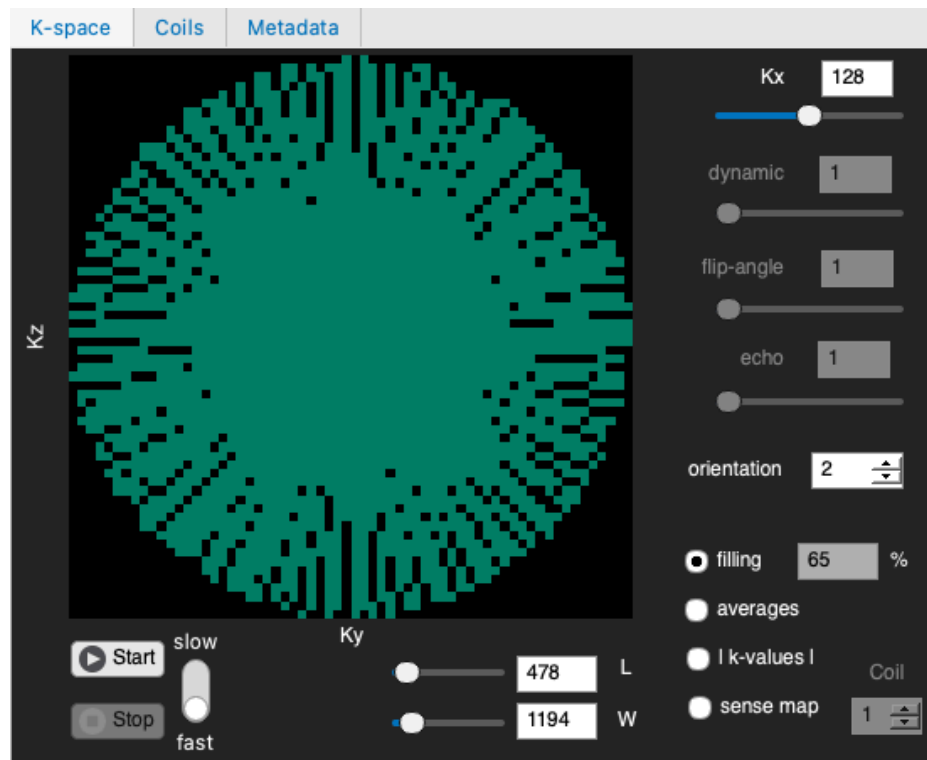
```
-48      (views   k-point 1)
0        (views2  k-point 1)
-47      (views   k-point 2)
0        (views2  k-point 2)
-46
1
...
```

3. **Undersampled 2D and 3D data** — pe1-order = 5 ("nrLUT"). The trajectories are read from the first .txt file in the folder of the MRD file (3D UTE: lut\*.txt).

The readout profile of 3D UTE data, `ktraj.txt` in the settings folder, has one column for all axes or three for x, y and z; the app writes its own when there is none. A comment line in it can give the data offset, which is then used instead of an estimated one (see Radial below). Fast spin echo scans with the scanner's `rtable.rtv` are read in its order.

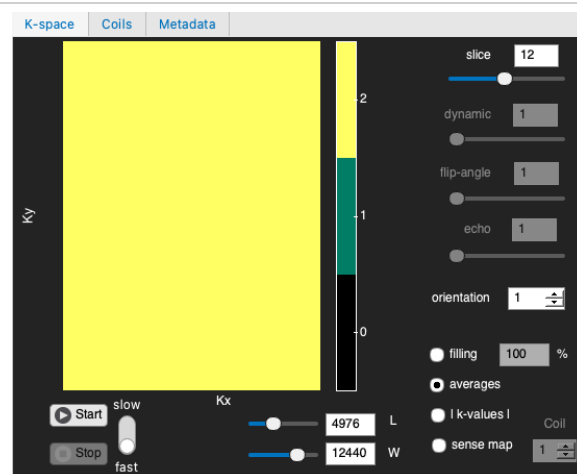
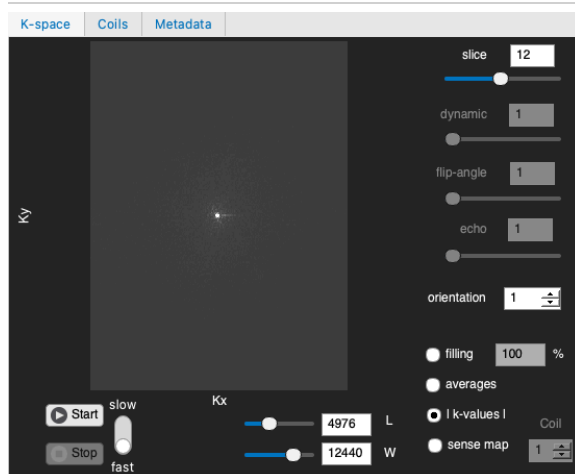
## Step 2: Inspecting k-space

The raw k-space data is displayed in the K-space tab of panel 2. The radio buttons show the acquired k-space points (**filling**), the averages per k-point (**averages**), the absolute value of k-space (**| k-values |**) or a coil sensitivity map (**sense map**). The percentage of k-space points acquired is shown beside **filling** (65 % for the dataset below).



## | k-values |

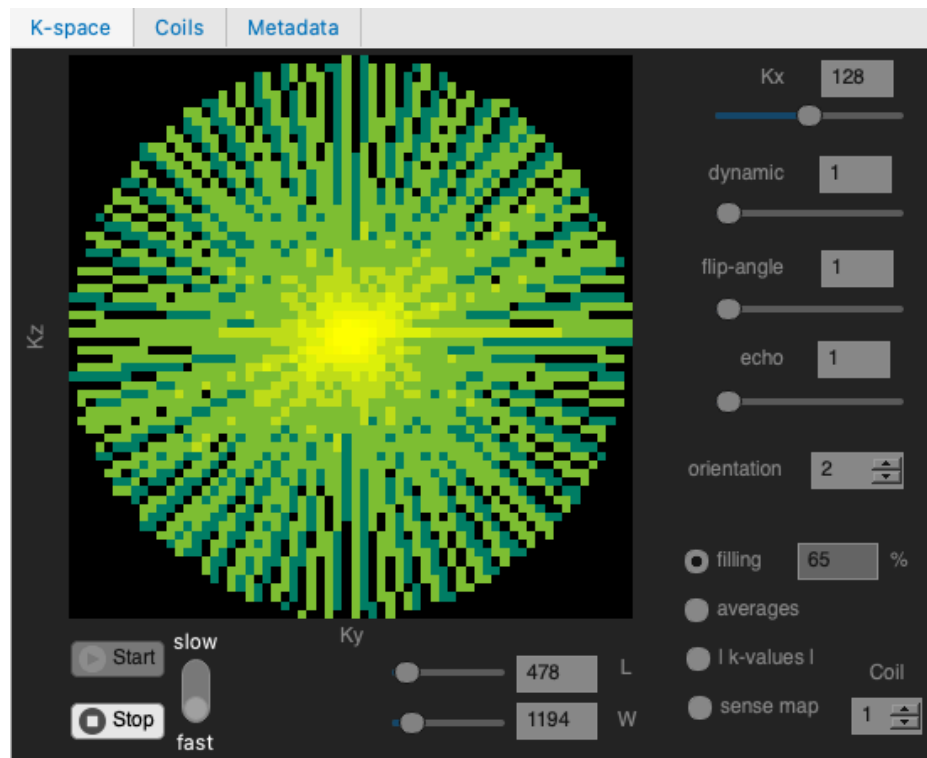
## averages



The sliders step through the third k-space dimension (**Kz**, or **slice** for 2D data), the repetitions (**dynamic**), the flip angles (**flip-angle**) and the echoes (**echo**), if applicable; for a series that steps through TR, TI or CINE frames the slider is named so. The **orientation** spinner chooses the k-space plane shown, the **Coil** spinner the coil, and **L** and **W** set the level and window.

### Step 3: The k-space trajectory

Press **Start** to start an animation of the k-space trajectory, and **Stop** to stop it. The switch beside them toggles between **fast** and **slow** animation.



#### Step 4: Fourier transform reconstruction

For a first quick inspection of the reconstructed images, press **Reco** with **CS reco** unticked: a Fourier transform reconstruction (for radial, UTE and ZTE data, a NUFFT or gridding).

**X**, **Y** and **Z** in the Dimensions panel set the size of the images: for Cartesian data k-space is zero-filled or cropped, for radial, UTE and ZTE data the images are resized. For 2D data the third field is **Slices** and keeps the number of slices acquired. **NR** sets the number of repetitions (see step 6).

#### Step 5: Coils

For acquisitions with a phased-array coil the Coils tab gives access to the channels: coils can be checked or unchecked to use or omit them in the reconstruction. Relative sensitivities can be set by hand, or determined with **auto sensitivity**, which for a CS reconstruction also computes the ESPIRiT coil maps shown under **Sensitivity maps**. Auto sensitivity is ticked at every load of multi-coil data.



| K-space                               | Coils                | Metadata                                             |
|---------------------------------------|----------------------|------------------------------------------------------|
| Coil selection                        |                      |                                                      |
| Coil                                  | Relative sensitivity | <input checked="" type="checkbox"/> auto sensitivity |
| <input checked="" type="checkbox"/> 1 | 1.000                | <input type="checkbox"/> 5 1.000                     |
| <input checked="" type="checkbox"/> 2 | 1.000                | <input type="checkbox"/> 6 1.000                     |
| <input type="checkbox"/> 3            | 1.000                | <input type="checkbox"/> 7 1.000                     |
| <input type="checkbox"/> 4            | 1.000                | <input type="checkbox"/> 8 1.000                     |
| Sensitivity maps                      |                      |                                                      |
| 1                                     | 2                    | 3                                                    |
| 4                                     | 5                    | 6                                                    |
| 7                                     | 8                    |                                                      |
| slice                                 |                      | 8                                                    |

## Step 6: Reconstruction

Ticking **CS reco** in the Dashboard gives access to the Compressed Sensing (CS) reconstruction parameters in the Reconstruction tab of panel 3.

| Scan parameters    | Geometry                   | Reconstruction                                       | Radial                           |
|--------------------|----------------------------|------------------------------------------------------|----------------------------------|
| Compressed sensing |                            | Corrections                                          | De-noising / filtering           |
| WVxyz              | 0.0010                     | distortion corr. <input checked="" type="checkbox"/> | PCA <input type="checkbox"/>     |
| TVxyz              | 0.0000                     | POCS <input checked="" type="checkbox"/>             | BMD <input type="checkbox"/>     |
| LLRxyz             | 0.0000                     | registration <input checked="" type="checkbox"/>     | de-ring <input type="checkbox"/> |
| TVtime             | 0.0000                     | FSE navigator <input checked="" type="checkbox"/>    | Tukey <input type="checkbox"/>   |
| CS iterations      | 500                        | FSE envelope <input type="checkbox"/>                |                                  |
| Sum acquisitions   |                            |                                                      |                                  |
| $\Sigma$ NR        | <input type="checkbox"/> 1 | SoS                                                  | <input type="button" value="v"/> |
| $\Sigma$ NE        | <input type="checkbox"/> 1 | SoS                                                  | <input type="button" value="v"/> |

The CS regularization parameters are:

1. **WVxyz** — wavelet regularization in the spatial (x, y and z) dimensions.
2. **TVxyz** — total variation constraint in the spatial dimensions.
3. **LLRxyz** — locally low-rank constraint in the spatial dimensions.
4. **TVtime** — total variation constraint in the dynamics (repetitions, NR) dimension.

The spatial dimensions of the reconstructed data can be increased or decreased. The number of repetitions (**NR**) can also be changed to undersample or average data in the dynamics dimension.



Pressing **Reco** starts the CS reconstruction, with BART when the app found it. Without BART, Cartesian, 2D radial and 3D UTE data are reconstructed with the app's own sigpy engine, with all four regularization terms and the ESPIRiT coil maps; ZTE data are then reconstructed without CS, and the CS checkbox is off for them.

For multi-slab 3D (MOTSA) scans, the slabs are put where they were acquired and blended over **slab overlap (pixels)**, in the Geometry tab, on either side of each junction.

## Further options in the Reconstruction tab

**CS iterations** sets the number of iterations of the CS reconstructions of 2D radial and 3D UTE data, and of ZTE data with BART. A running reconstruction can be stopped with **abort** in the Dashboard.

**De-noising / filtering** is applied to the reconstructed images; the methods can be combined.

- **PCA** removes noise by principal component analysis with the Marchenko–Pastur criterion, over the images of each voxel in a series (dynamics, diffusion directions, flip angles, echoes), in a window chosen from the data; for EPI it runs before the EPI corrections.
- **BMD** applies block-matching denoising: BM3D per slice for 2D data, BM4D per volume for 3D, UTE and ZTE data.
- **de-ring** suppresses Gibbs ringing. It is not available for radial, UTE and ZTE data. It works on the complex images where it can, and on the magnitude when the echo lies more than two samples off the k-space center, where the complex images' suppression removes little.

## Corrections.

- **Tukey** tapers the edges of k-space against ringing, at some cost in resolution; loading a Cartesian scan unticks it, a radial or UTE scan ticks it.
- **FSE navigator** (ticked by default) corrects the phase and readout shift of each echo of a fast spin echo: in 2D by its navigator train, where they are large enough to matter; in 3D per image slice, from the neighboring slices, where that lowers the slice background.
- **FSE envelope** smooths the amplitude steps between the echoes of a fast spin echo, measured by its navigator: less ghosting at the same signal-to-noise ratio, slightly less sharp.

The FSE boxes are available for fast spin echo scans with a navigator.

**Sum acquisitions.** With **Σ NR** or **Σ NE** ticked, the dynamics or the echoes are combined into one image by **SoS**, **Max** or **Sum**; the field beside the box selects which, with a 1 or 0 for each.

## EPI scans

2D EPI scans, single-shot or read-interleaved, spin echo or gradient echo, with or without partial Fourier, are reconstructed as diffusion series (DTI, several b-values) or fMRI time series. Ghost correction is always applied; drift correction is applied when the series has volumes without diffusion weighting to measure it on; and the images are masked when the study has a mask scan of the same geometry (named in the scan settings, or the nearest one). Three corrections in the EPI part of the tab are on by default:

- **distortion corr.** corrects the geometric distortion with a partner scan of the same study, with the same slices and echo train and the opposite phase-encoding direction, which the app finds by

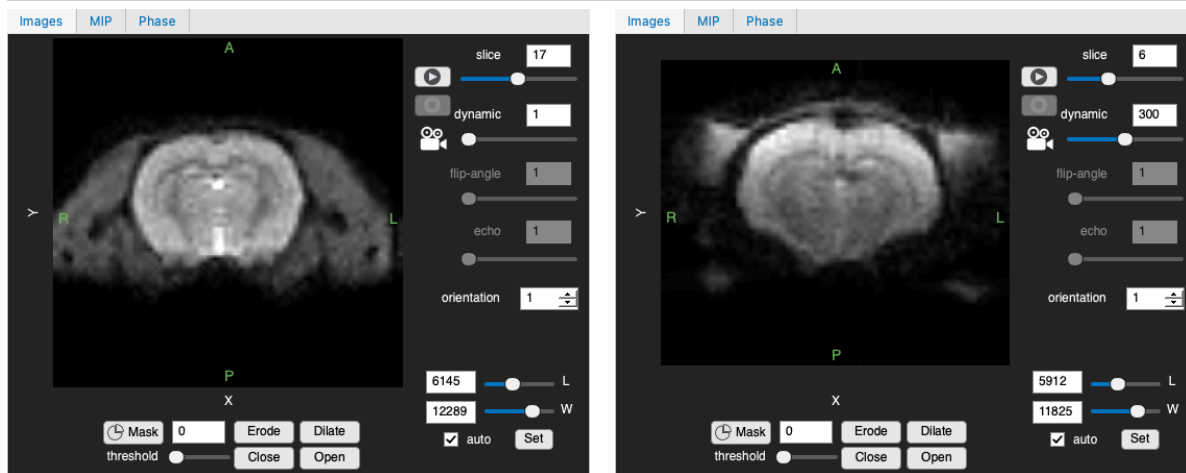


itself: a displacement field fitted to the physical distortion model is applied to every volume. Without a partner scan the message box says so and nothing is changed.

- **POCS** completes a partial Fourier scan: sharper images, at some cost in signal-to-noise ratio.
- **registration** aligns the volumes with the mean  $b = 0$  volume (diffusion) or the first volume (fMRI), by shifts in the image plane.

| Scan parameters                                                            | Geometry                 | Reconstruction                                       | Radial                           |
|----------------------------------------------------------------------------|--------------------------|------------------------------------------------------|----------------------------------|
| <div>Compressed sensing      Corrections      De-noising / filtering</div> |                          |                                                      |                                  |
| WVxyz                                                                      | 0.0010                   | distortion corr. <input checked="" type="checkbox"/> | PCA <input type="checkbox"/>     |
| TVxyz                                                                      | 0.0000                   | POCS <input checked="" type="checkbox"/>             | BMD <input type="checkbox"/>     |
| LLRxyz                                                                     | 0.0000                   | registration <input checked="" type="checkbox"/>     | de-ring <input type="checkbox"/> |
| TVtime                                                                     | 0.0000                   | FSE navigator <input checked="" type="checkbox"/>    | Tukey <input type="checkbox"/>   |
| CS iterations                                                              | 500                      | FSE envelope <input type="checkbox"/>                |                                  |
| Sum acquisitions                                                           |                          |                                                      |                                  |
| $\Sigma$ NR                                                                | <input type="checkbox"/> | 1 1 1 1 1 1 1 1 1 1 1 1                              | SoS <input type="text"/>         |
| $\Sigma$ NE                                                                | <input type="checkbox"/> | 1                                                    | SoS <input type="text"/>         |

**Diffusion (DTI, 130 volumes): a  $b = 0$  volume      fMRI (600 dynamics): slice 6, dynamic 300**



The **DCM / NIFTI** export of an EPI series also writes a BIDS dataset in `BIDS/sub-<study>/ses-1: func` (fMRI) or `dwi` with the gradient table (diffusion), and `anat`, a fast spin echo scan of the study that the app finds and reconstructs itself (T2w for a TE above 30 ms, otherwise T1w; the best scan of the other weighting is included as well). An fMRI series needs this scan, a diffusion series does not.

For a diffusion scan with at least 6 directions and a  $b = 0$  volume, MRtrix3 and DSI Studio scripts for tractography (`tractography_MRtrix_<tag>` and `tractography_DSISudio_<tag>`, `.sh` and `.bat`) are written into the NIFTI folder; the app does not run them.

### Reconstruction of radial scans

The Radial tab gives access to advanced reconstruction parameters for 2D radial, 3D UTE and ZTE scans.

| Scan parameters                                                                                                                          | Geometry | Reconstruction                                                                                                                                                                                                                                           | Radial |
|------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>Gradient delays</b>                                                                                                                   |          | <b>Corrections</b>                                                                                                                                                                                                                                       |        |
| calibration <input checked="" type="checkbox"/> <span>abort</span><br>ring method <input checked="" type="checkbox"/>                    |          | center echo <input checked="" type="checkbox"/><br>phase correction <input checked="" type="checkbox"/><br>amplitude correction <input type="checkbox"/><br>density correction <input checked="" type="checkbox"/><br>large FOV <input type="checkbox"/> |        |
| <b>ZTE</b>                                                                                                                               |          | dead time ( $\mu$ s) 11.5<br>shift (samples) 4<br>full echo <input type="checkbox"/><br>registration <input type="checkbox"/>                                                                                                                            |        |
| Gx delay 0.00000<br>Gy delay 0.00000<br>Gz delay 0.00000<br>Sxy delay 0.00000<br>Sxz delay 0.00000<br>Syz delay 0.00000<br>data offset 4 |          |                                                                                                                                                                                                                                                          |        |

Tick **calibration** to estimate the gradient delays from the data at the next reconstruction; the delay fields are then filled in and the checkbox is unticked. 2D radial scans load with calibration and **ring method** ticked. With ring method ticked the delays are estimated with BART's `estdelay`, using the method of Block and Uecker; RING, after which the checkbox is named, is not used, as it is unreliable on 0–180° trajectories. Without BART, ring method takes the delays from the positions of the echoes in the spokes instead. Otherwise, and for 3D UTE, an iterative low-rank method is used, which is much slower and does not need BART; **abort** stops it. For 3D UTE the calibration also estimates **data offset**, the number of samples dropped from the start of each readout. The delays, in samples, are **Gx**, **Gy** and **Sxy** for 2D radial data, and **Gx**, **Gy**, **Gz**, **Sxy**, **Sxz** and **Syz** for UTE and ZTE; they are reset when a data set is loaded.

The Images tab below shows a reconstructed 2D radial scan of a phantom.



**center echo** centers the echoes before reconstruction and **phase correction** corrects their phase; a 3D UTE scan loads with both unticked and data offset 8.

**large FOV** reconstructs a 3D UTE scan over 1.5 times the field of view and crops it, so that less signal leaks in from beyond the field of view; it takes about a quarter longer. It applies to the NUFFT reconstruction, not to CS.

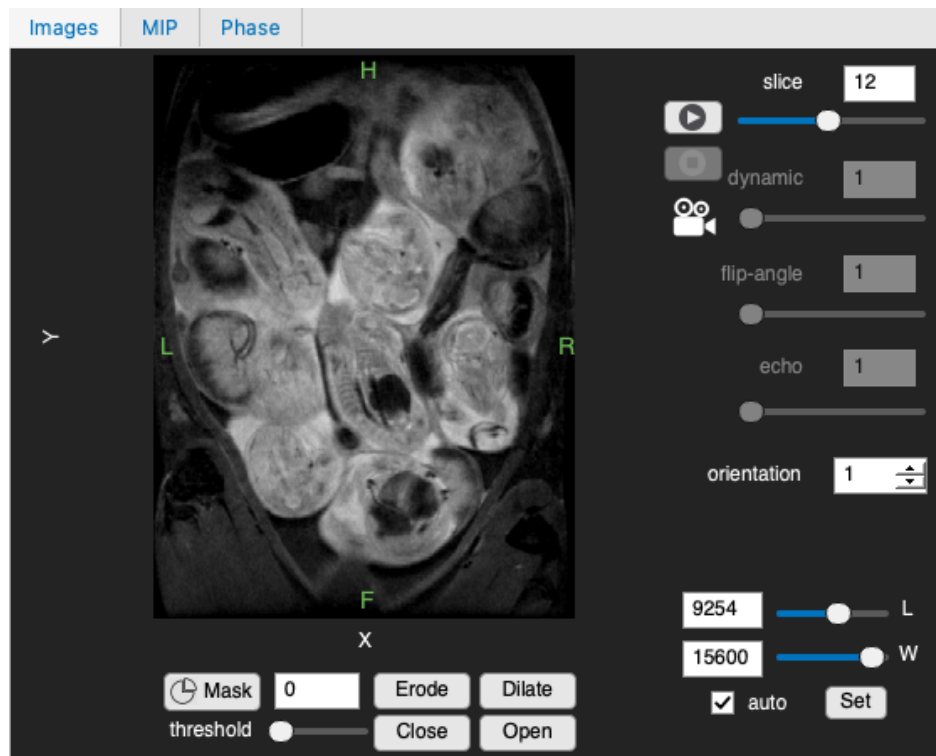
**density correction** performs a 2D or 3D density correction in k-space to account for non-uniform sampling. It only applies to CS reconstructions, and its checkbox is on only with **CS** ticked.

**amplitude correction** scales every spoke to the mean peak amplitude. The Tukey filter, which tapers the readout, is set in the Reconstruction tab.

**ZTE** scans have no calibration and none of these corrections. Their **dead time ( $\mu$ s)** is the delay between the excitation and the first sample, 11.5  $\mu$ s by default, and **shift (samples)** the first sample kept, 4 by default; changing shift rebuilds k-space. With **full echo** both halves of the echo are used, and **registration** aligns the dynamics of a series in 3D.

## Step 7: Viewing the reconstructed images

After reconstruction the MR images are displayed in panel 4.



The image orientation can be changed with the **orientation** spinner; the green letters around the image give the anatomical directions. The MIP and Phase tabs follow it.

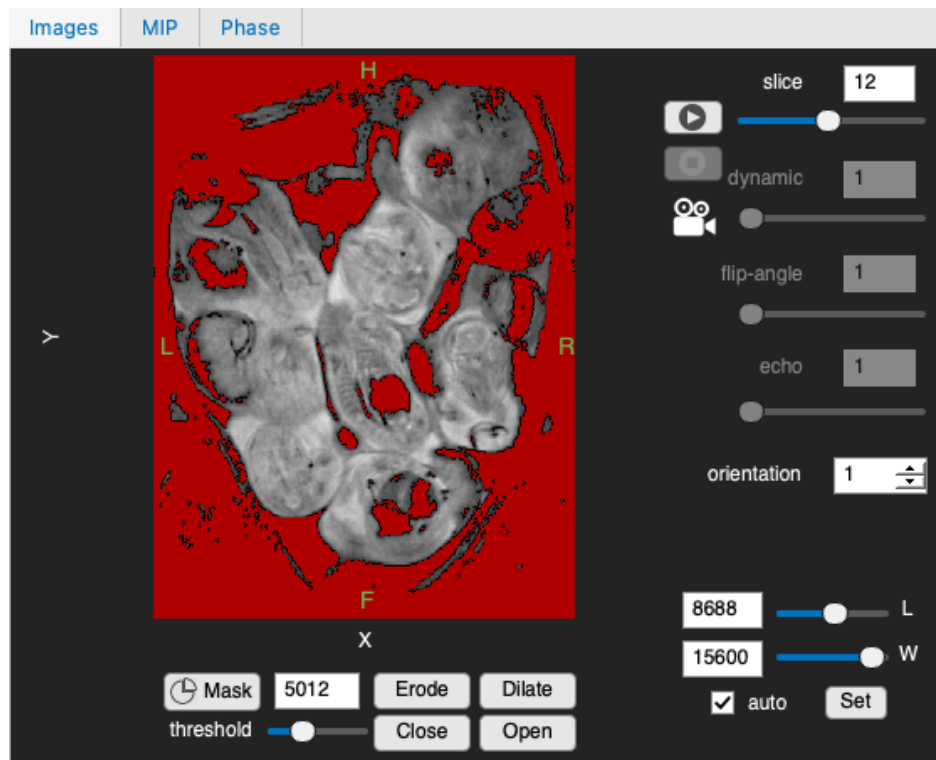
Image intensity can be adjusted with the level (**L**) and window (**W**) sliders. Checking **auto** adjusts the intensity automatically; **Set** takes the level and window of the image shown and unticks auto.

Press the play button to start a movie of the images as a function of the dimension shown next to the camera icon (for example the dynamics), and the stop button to stop it.



When the pointer is over the image, a small toolbar appears: its zoom button zooms with a box, a click or the wheel, and its home button (**Whole image**) shows the whole image again. With zoom off, the scroll wheel steps through the slices.

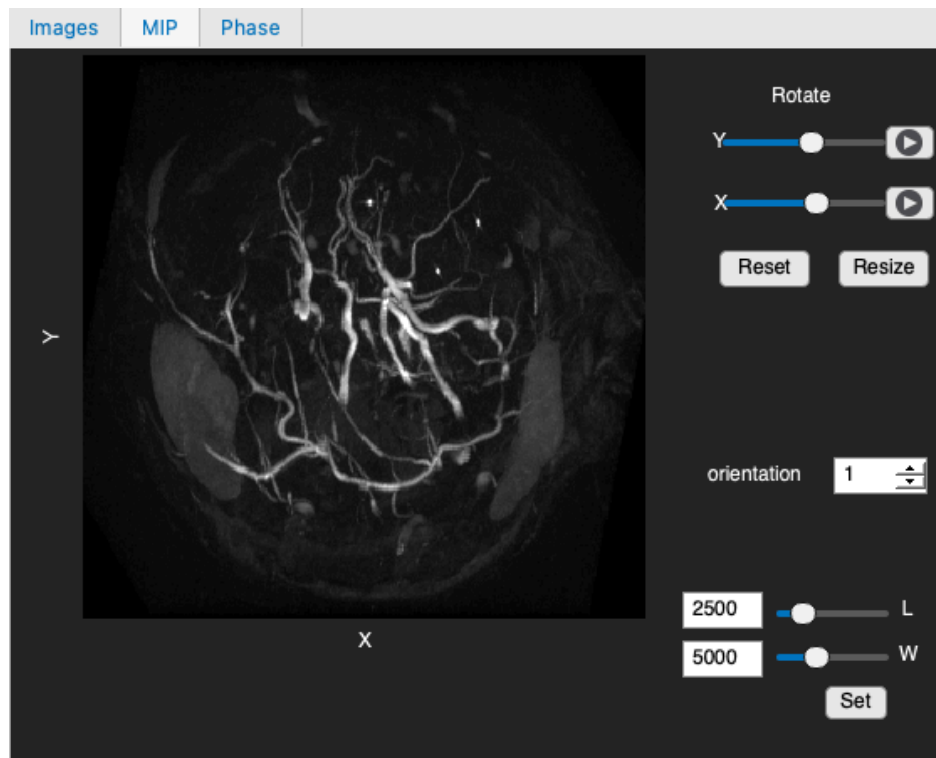
Images can be segmented (on the basis of a threshold) by pressing **Mask**. Use the **threshold** slider or the morphological operations (**Erode**, **Dilate**, **Open**, **Close**) to fine-tune the segmentation. The masked-out pixels are shown in red. The mask is used for display, the MIP and phase unwrapping only.



## Maximum intensity projection (MIP)

In the MIP tab a maximum intensity projection is shown. The MIP can be turned with the **X** and **Y** sliders under **Rotate**, and animated with the play buttons beside them, 5° per step (pressing again stops). **Reset** stops the movies and returns to the start; **Resize** halves the volume for speed. The MIP has its own level, window and **Set**.

The object keeps its size as it turns: the view is the same at every angle, and the corners of a turned projection that reach beyond it are cut off.



## Phase images

The Phase tab shows the phase images, with their own sliders, movie and orientation. The phase can be unwrapped by pressing the **Unwrap** button; pixels outside the mask keep their wrapped value.

### Phase

### Unwrapped



## Step 8: Metadata

The **Metadata** tab, beside the K-space and Coils tabs, holds the record of the scan, which the DCM / NIFTI and GIF exports write with the images. Its fields stand in three columns:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;



- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.

| K-space                                                                | Coils                       | Metadata                |
|------------------------------------------------------------------------|-----------------------------|-------------------------|
| <div> <div>Subject</div> <div>Session</div> <div>Contrast</div> </div> |                             |                         |
| animal ID 882827                                                       | project Stroke pilots       | agent e.g. gadob...     |
| species mouse                                                          | license ethics approva...   | conc. (mM) e.g. 100     |
| strain C57BL/6J                                                        | date, time 2023-01-13 11:3' | dose (mmol/kg) e.g. 0.3 |
| strain RRID e.g. IMSR_JA...                                            | operator name               | volume (μL) e.g. 75     |
| sex female                                                             | institution AMC             | route choose            |
| age (weeks) e.g. 12                                                    | scanner MRS7000 7 T 10      | inj. dynamic e.g. 20    |
| weight (g) 23.1                                                        | anesthesia isoflurane 2%    |                         |
| group stroke                                                           | heart (bpm) e.g. 450        |                         |
| time point e.g. D0                                                     | resp. (/min) e.g. 50        |                         |
|                                                                        | temp. (°C) e.g. 37.0        |                         |
|                                                                        | notes                       |                         |

At a load the program fills in what the scan's files say: from the .rpr file beside the MRD file the animal ID and the time point (from the patient name typed at the console: 20240212\_926752\_D0 is animal 926752, day 0), the date and time, the institution and the scanner, and from the scanner's DICOM images, where they are found, the project. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty.

What was filled in for an earlier scan of the same exam is filled in again, from the records the apps wrote in the exam's Metadata folder (P2ROUD, Retrospective and the mapping apps alike) and from the tab as it was for the scan loaded before: the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature and the notes belong to one scan and are never taken from another.

A value the program filled in is shown in blue; every field can be corrected, and then counts as yours. Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

## Step 9: Image export

There are three ways to export the images, in the Export panel of the Dashboard.

1. **DCM / NIFTI** — exports the images as DICOM files for further processing in third-party software. When the scanner's own DICOM images are found, their headers are used; otherwise the headers are made from the MRD data, with the orientation taken from the scan angles and the slices centered; the patient, institution and scanner in them are those of the scan's .rpr file, and stay empty where it has none. The DICOM files carry the species, strain, sex, age and weight of the animal where the Metadata tab has them. The images are also written as NIfTI files with their



geometry (with dcm2niix), and for diffusion scans with .bval and .bvec files; recoparameters\_<tag>.txt records the scan and the settings used. For EPI series a BIDS dataset and scripts for tractography are written as well (see "EPI scans"). When the scanner's DICOM export of the scan is found in .../Data/<study>/DICOM/<scan>, the files go into .../Data/<study>/DICOM/<series>P/<n> and Nifti/<tag>P/<n>, a new numbered folder <n> each time; otherwise into a folder that is asked for (DICOM/<tag>P/<n> there).

2. **GIF** — the data are exported as (animated) GIF files, one per slice, flip angle or echo. When the MRD file lies in .../MRD/<study>/<scan>, they go into .../Data/<study>/GIF/<tag>P/<n>, a new numbered folder each time; otherwise into a folder that is asked for.
3. **MRD** — the data are stored as a new MRD file, p2roud\_<n>\_<tag>.MRD, in the folder of the input MRD file (or in a folder asked for, when that cannot be written), with recoparameters\_<tag>.txt and, if the program finds an input .rpr file, a new .rpr file. The scan's rtable.rtv is renamed origtable.rtv.orig. Additionally, when the program runs on the scanner computer (Windows), the data are reconstructed by the scanner software and new SUR files, and for nrLUT scans DICOM images made from them, are placed in the data directory for viewing and planning of next scans in the MR Solutions preclinical software. nrLUT scans are exported to MRD by themselves after every reconstruction.

The DCM / NIFTI and GIF exports also write, in .../Data/<study>/Metadata/<tag>P/<n> (or under the folder asked for):

- metadata.json — the Metadata tab's record;
- p2roud-report\_<tag>.json and .txt — the report of the export: the program and its libraries, the scan and a checksum of each of its files, the steps of the session, the scan and reconstruction settings, and every file the export wrote with its checksum;
- ro-crate-metadata.json — the same as an RO-Crate (<https://www.researchobject.org/ro-crate/>), a standard description of the folder for data repositories.

The paths in them are relative to the study folder, so they hold no names of your own folders.

## Step 10: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while the program is busy: the window disappears, and the program closes as soon as the running step has finished. Nothing the program started (BART) keeps running after it.



## 7. Credits

- **BART toolbox and compressed sensing algorithms** — <https://mrirecon.github.io/bart/>. Michael Lustig, David Donoho, John M. Pauly, Sparse MRI: The application of compressed sensing for rapid MR imaging, Magn Reson Med 58, 1182–1195 (2007).
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- **NUFFT** — <https://web.eecs.umich.edu/~fessler/>
- **Phase unwrapping** — M. A. Herráez, D. R. Burton, M. J. Lalor, and M. A. Gdeisat, "Fast two-dimensional phase-unwrapping algorithm based on sorting by reliability following a noncontinuous path", Applied Optics, Vol. 41, Issue 35, pp. 7437–7444 (2002); H. Abdul-Rahman, M. Gdeisat, D. Burton, M. Lalor, "Fast three-dimensional phase-unwrapping algorithm based on sorting by reliability following a non-continuous path", Proc. SPIE 5856, Optical Measurement Systems for Industrial Inspection IV, 32 (2005).
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  - M. Maggioni, V. Katkovnik, K. Egiazarian, A. Foi, "Nonlocal Transform-Domain Filter for Volumetric Data Denoising and Reconstruction", IEEE Transactions on Image Processing, vol. 22, pp. 119–133 (2013).
  - Y. Mäkinen, S. Marchesini, A. Foi, "Ring Artifact and Poisson Noise Attenuation via Volumetric Multiscale Nonlocal Collaborative Filtering of Spatially Correlated Noise", Journal of Synchrotron Radiation, vol. 29, pp. 829–842 (2022).
- **NIfTI conversion** — dcm2niix: C. Rorden et al., <https://github.com/rordenlab/dcm2niix>.